

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

MATHEMATICS

4008/1, 4028/1

PAPER 1

JUNE 2008 SESSION

2 hours 30 minutes

Candidates answer on the question paper.

Additional materials:

Geometrical instruments

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer all questions.

Write your answers in the spaces provided on the question paper.

If working is needed for any question it must be shown in the space below that question.

Omission of essential working will result in loss of marks

Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise.

Mathematical tables, slide rules and calculators should not be brought into the examination room.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

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NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR
CALCULATORS MAY BE USED IN THIS PAPER

For
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Use

- 1 (a) Express 0,000 0526 in standard form.
- (b) Estimate by first rounding off each number to one significant figure,
the value of $\frac{3,04 \times 1,98 - 9,48}{2,91}$.

Answer

(a)

(b)

2 Express

- (a) $5,3 \text{ km}^2$ in hectares,
(b) 20 litres in cubic centimetres,
(c) 408 hours in weeks and days.

Answer (a) _____ ha [1]
(b) _____ cm^3 [1]
(c) _____ weeks _____ days [1]

- 3 (a) Find the value of the base y such that $6703_y - 725_y = 5756_y$.
(b) Convert 134_5 to a number in base two.

Answer (a) _____ [1]
(b) _____ [2]

4 In a shop one loaf of bread cost \$65 000.

(a) Find

(i) the cost of a dozen loaves of bread,

(ii) the number of loaves of bread that could be bought for \$4 550 000.

(b) Write down the type of variation which connects the cost of bread and the number of loaves of bread bought.

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Answer (a) (i) \$_____ [1]
(ii) _____ [1]
(b) _____ variation [1]

- 5 (a) Write down **one** capital letter of the alphabet which has rotational symmetry of order 2 but no line of symmetry.
- (b) The mean of m , p , q and r is 8.25.
The mean of m , n , p , q and r is 4.6.
Find the value of n .

Answer (a) _____ [1]

(b) _____ [2]

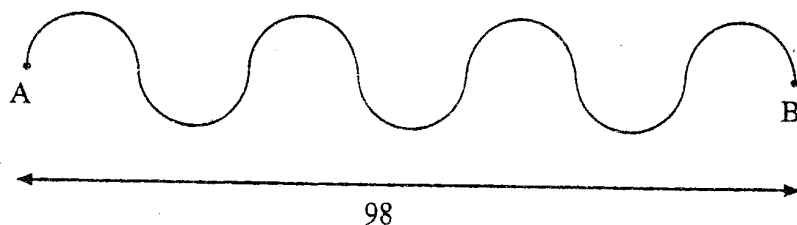
- 6 The radius, r cm, of a circle is given as 9 cm, correct to the nearest whole number.

- (a) Write down the limits between which r must lie.
- (b) Calculate the least possible circumference of the circle, giving your answer in terms of π .

Answer (a) _____ [1]

(b) _____ cm [2]

6



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Use

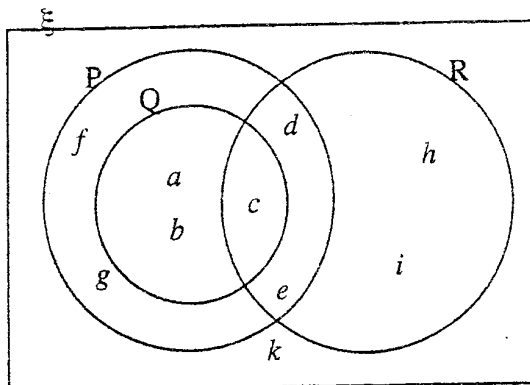
The diagram shows a pattern which is made up of identical semi-circular arcs. The shortest distance between the ends A and B is 98 cm.

Calculate

- (a) the diameter of each semi-circle,
- (b) the distance from A to B along the curves.

$$\left(\text{Take } \pi = \frac{22}{7} \right)$$

Answer (a) _____ cm [1]
(b) _____ cm [2]



The Venn diagram shows the universal set, ξ and the sets P, Q and R with their members.

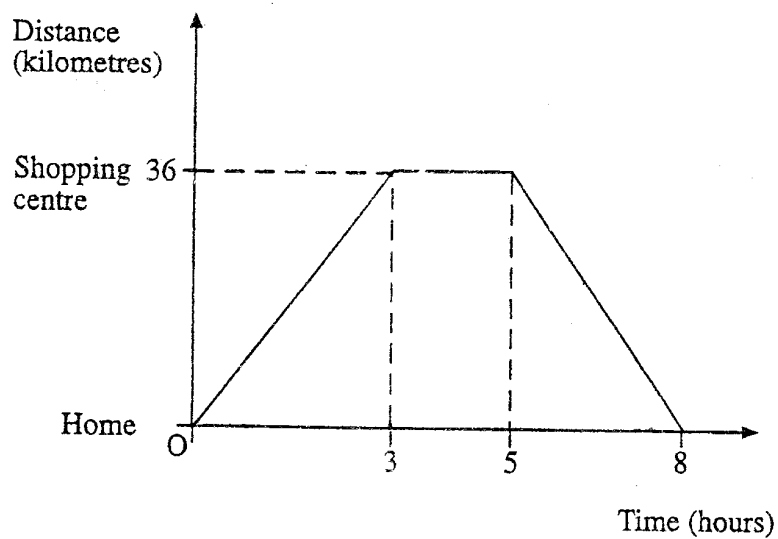
(a) From the venn diagram, list the members of

(i) $P \cap Q$,

(ii) $P' \cap Q'$,

(b) Find $n(P)$.

Answer (a) (i) $P \cap Q =$ _____ [1]
 (ii) $P' \cap Q' =$ _____ [1]
 (b) $n(P) =$ _____ [1]



The diagram is the distance-time graph of a cyclist who goes out shopping and returns home.

Calculate

- (a) the time spent at the shopping centre,
- (b) the average speed for the whole journey.

Answer (a) _____ hours [1]
(b) _____ km/h [2]

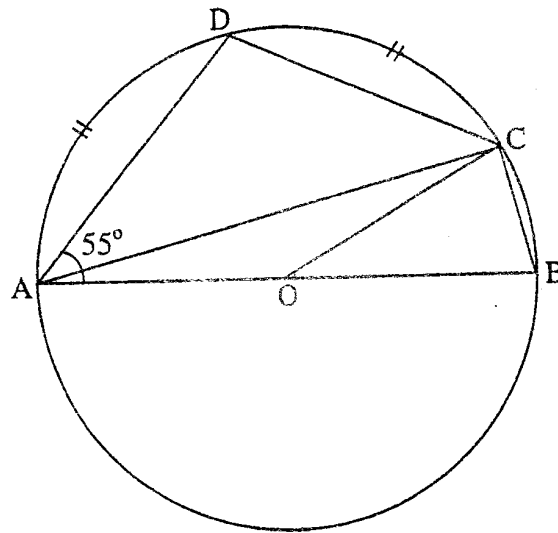
- 10 A bag contains 60 identical marbles, except for colour. In the bag 36 are green, x are red and the rest are blue.

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- (a) If the probability of selecting a red marble at random is $\frac{1}{3}$, find x .
- (b) Two marbles are drawn, one after the other, with replacement. Find the probability that neither marble is green, giving your answer as a common fraction in its lowest terms.

Answer (a) $x =$ _____ [1]

(b) _____ [2]



In the diagram ABCD is a circle centre O. Arc AD = arc DC, $\angle DAO = 55^\circ$ and AOB is a straight line.

Calculate

- (a) $\angle BCD$,
- (b) $\angle ACB$,
- (c) $\angle DAC$.

- Answer
- (a) $\angle BCD =$ _____ [1]
 - (b) $\angle ACB =$ _____ [1]
 - (c) $\angle DAC =$ _____ [1]

12 Evaluate

(a) $(-32)^{\frac{1}{5}}$,

(b) $\frac{1}{5^{-3}}$.

Answer (a) _____ [2]
(b) _____ [2]

13 Factorise completely

(a) $15m + 18 - 10mn - 12n$,

(b) $27x^2 - 12y^2$.

Answer (a) _____ [2]
(b) _____ [2]

- 14 The following is an extract of Mr Dube's telephone bill for the month of January 2002 whereby x is the number of units used and y is the Value Added Tax (VAT).

DATE	ACCOUNT DETAILS	AMOUNT
31 Jan 2002	Rental from 01/02/02 to 28/02/02	140,00
	Metered units 02781 01685 (x units)	2 192,00
	Sub Total	2 332,00
	VAT @ 15,0%	y
	Amount Due	

- (a) Find,
- (i) the units used, x
- (ii) the cost per unit.
- (b) Calculate the Value Added Tax, y .

Answer (a) (i) _____ [1]

(ii) \$ _____ [2]

(b) \$ _____ [1]

- 15 (a) A triangle has sides of length 7 cm, 15 cm and x cm. If x is an integer, find
- (i) the minimum value of x ,
 - (ii) the maximum value of x .
- (b) The sum of the interior angles of an irregular polygon is 5040° . Find the number of sides of the polygon.

Answer (a) (i) minimum $x =$ _____ [1]
(ii) maximum $x =$ _____ [1]
(b) _____ [2]

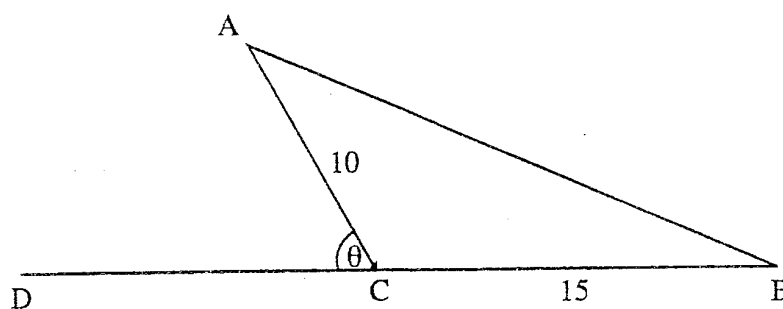
16 The selling price of school shoes at a departmental store in January 2005 was \$750 000 per pair.

- (a) Calculate the amount paid for one pair of school shoes if a 10% discount was allowed.
- (b) In February 2005 the price rose by 15%. Calculate the new selling price of one pair of school shoes.

Answer (a) \$ _____ [2]

(b) \$ _____ [2]

17.



In the diagram DCB is a straight line. $BC = 15$ cm, $AC = 10$ cm and $\hat{ACD} = \theta$.

Given that $\sin \theta = \frac{3}{5}$, find

- (a) $\cos \theta$,
- (b) the area of $\triangle ABC$.

Answer (a) $\cos \theta =$ _____ [2]

(b) _____ cm^2 [2]

- 18 Bulawayo and Harare are 400 km apart by air.

A helicopter leaves Bulawayo at 2.30 p.m. and arrives in Harare at 4.10 p.m. on the same day. Calculate

- (a) the average speed of the helicopter,
- (b) the arrival time in Bulawayo in the 24-hour clock on the return journey if the helicopter leaves Harare at 10.36 p.m. and takes 1 hour 49 minutes.

Answer (a) _____ km/h [2]

(b) _____ [2]

19. It is given that $\mathbf{p} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -15 \\ y \end{pmatrix}$.

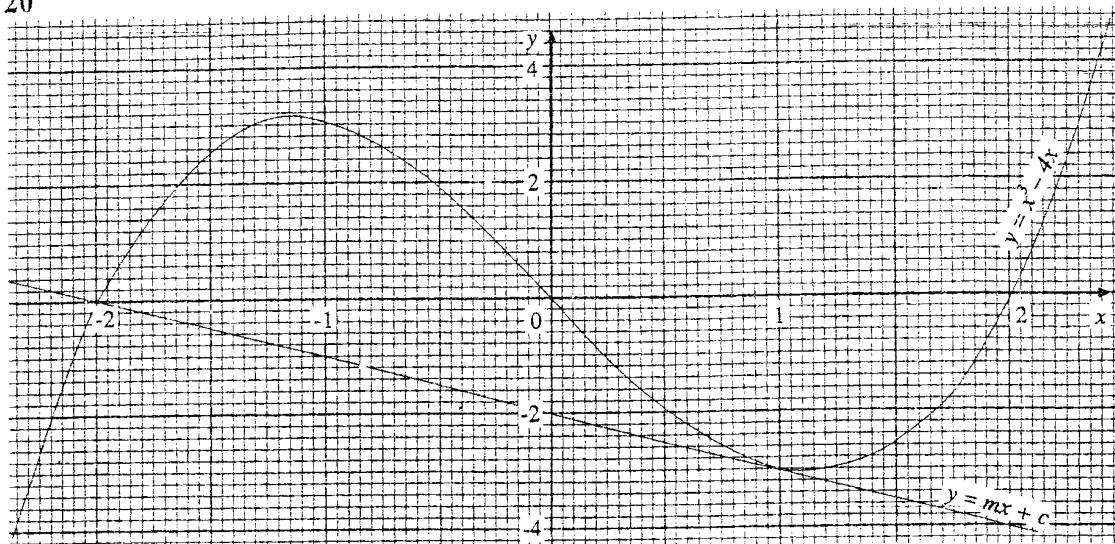
- (a) Calculate $|\mathbf{p}|$, leaving your answer in surd form.
- (b) If $\mathbf{p}$ is parallel to $\mathbf{q}$, calculate the value of y .

For
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Use

Answer (a) _____ [2]

(b) _____ [2]

20



In the diagram the line $y = mx + c$ touches the curve $y = x^3 - 4x$ at $(1; -3)$.

Write down

- (a) the value of c ,
- (b) the value of m ,
- (c) the gradient of $y = x^3 - 4x$ when $x = 1$,
- (d) the range of values of x for which $mx + c > x^3 - 4x$.

- Answer
- (a) $c =$ _____ [1]
 - (b) $m =$ _____ [1]
 - (c) _____ [1]
 - (d) _____ [1]

21 It is given that $M = \begin{pmatrix} 2x & x \\ x & 2x \end{pmatrix}$.

- (a) Find M^2 in terms of x .
- (b) Find x given also that $|M| = 48$.

Answer (a) $M^2 =$ _____ [2]

(b) $x =$ _____ or _____ [3]

22 It is given that y varies inversely as $x + 4$.

- (a) Express y in terms of x and a constant k .
- (b) Given also that $x = 6$ when $y = 2$, find the value of k .
- (c) Find the value x when $y = 8$.

Answer (a) $y =$ _____ [1]

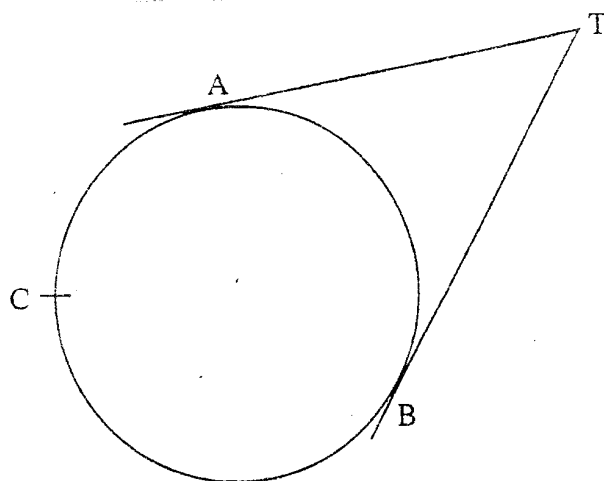
(b) $k =$ _____ [2]

(c) $x =$ _____ [2]

- 23 (a) At a certain party the ratio *men: women: children* was 5: 6: 4. If there were 80 children at the party, calculate the total number of people at the party.
- (b) The area of a certain country is given as $8,5 \times 10^6 \text{ km}^2$. The area of the country's map is given as $3,4 \times 10^5 \text{ cm}^2$. Find, in the form 1: n, the scale used in drawing the map.

Answer (a) _____ [2]

(b) _____ [3]



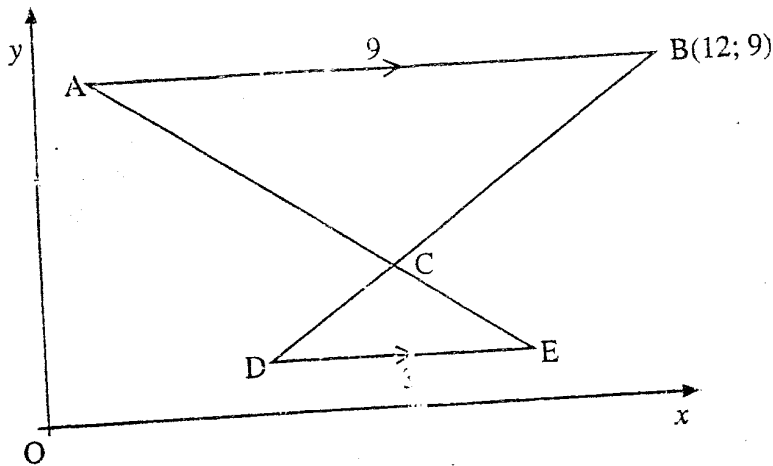
In the diagram TA and TB are tangents to the circle ABC.

- (a) Using a ruler and compasses only, construct
- (i) the locus of points equidistant from A and C,
 - (ii) the locus of points equidistant from AT and BT.
- (b) Write down the radius of the circle.

Answer

(a)	(i)	on diagram.	[2]
	(ii)	on diagram.	[2]
(b)		_____ cm	[1]

25



In the diagram ACE and BCD are straight lines. AB is parallel to DE , $AB = 9$ cm, $BC = 3$ cm and B is the point $(12; 9)$.

- (a) Describe **fully** the **single** transformation which maps $\triangle ABC$ onto $\triangle EDC$.
- (b) $\triangle ABC$ is mapped onto $\triangle A'B'C'$ by a transformation represented by the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. Calculate the coordinates of B' .

Answer

(a)

_____ [3]

(b)

(_____ ; _____) [2]

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26 Given that $\log 2 = 0,3010$ and $\log 3 = 0,4771$, calculate

- (a) $\log 6$,
- (b) $\log 1,5$,
- (c) $\log \sqrt{2}$.

Answer

(a) $\log 6 =$ _____ [2]

(b) $\log 1,5 =$ _____ [2]

(c) $\log \sqrt{2} =$ _____ [2]

X 45
D